

江西海瑞天然植物有限公司

JIANGXI HAIRUI NATURAL PLANT CO. LTD.

ADD: No.267 JiaHua Avenue,Jinggangshan Economic Development Zone, Ji'an County,Ji'an City, Jiangxi
Zip code:343100 Tel: 0796-8507999 FAX: 0796-8108396
Website: www.jxhairui.com

MATERIAL SAFETY DATA SHEET OF CLOVE OIL

1 CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

Product Name: Clove Oil

Manufacture: Jiangxi Hairui Natural Plant Co. Ltd.

Address: No.267 JiaHua Avenue, Jinggangshan Economic Development Zone, Ji'an County, Ji'an City, Jiangxi , P.R.C

Post code: 343100

Fax: 86-0796-8507999

Tel: 86-0796-8107398

2. COMPOSITION/INGREDIENT INFORMATION

Ingredients: Clove oil is the volatile oil obtained by the steam distillation of the

fresh leaves of clove
HS CODE: 3301299999

CAS #: 8000-34-8

3. HAZARDS IDENTIFICATION

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Xn Harmful

Routes of Entry

Eye Contact: Irritation will occur with possible damage

Skin Contact: May cause irritation or allergic dermatitis upon prolonged exposure

Inhalation: Inhalation may cause irritation in throat and lungs. High concentrations of vapor may cause anesthetic effects.

Ingestion: Harmful if swallowed

4. FIRST AID MEASURES

Eyes: Flush with plenty of water or eye wash solution for 15 minutes. Get medical attention if irritation persists.

Skin: Remove contaminated clothing. Wash with soap and flush with plenty of water. Seek medical attention if irritation persists.

Ingestion: Administer water or milk to dilute. Seek medical attention immediately or call a poison control center.

Inhalation: Avoid gross inhalation of fumes. Remove to fresh air and seek medical attention if needed.

5. FIRE FIGHTING MEASURES

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EXTINGUISHING MEDIA: Water vapor distillation distillation .

RECOMMEND: Extinguishing media to base of flames. Do not use direct water jet on burning material.

SPECIAL MEASURES: Avoid the vapour inhalation. Keep away from sources of ignition. Do not smoke. Wear positive pressure self-contained breathing apparatus & protective clothing.

EXTINGUISHING PROCEDURES: Close containers which may build up pressure when exposed to heat and should be cooled with water spray.

6. Special Firefighting Procedures:

Fire fighters should wear positive pressure self-contained breathing apparatus. Toxic vapors may be released. Cool containers exposed to flame with water. Spray extinguishing media directly into base of fire.

Unusual Fire & Explosion Hazards: None known

7. ACCIDENTAL RELEASE MEASURES (STEPS FOR SPILLS)

Personal Protection:

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Avoid ingestion

Follow good industrial practice as to housekeeping, personal hygiene and avoidance of prolonged or repeated contact with the skin or eyes.

Wear eye protection, gloves and aprons. Ventilate area well or wear approved respirators.

Methods for Cleaning Up:

Eliminate all sources of ignition

Ventilate area or wear approved respirators

Prevent liquid from entering sewers

Small spills can be wiped up with paper rags (place in closed metal waste container)

Large spills can be absorbed onto non-flammable absorbent.

Wash area with soap and water.

8. HANDLING AND STORAGE

PRECAUTIONS IN HANDLING: Apply good manufacturing practice & industrial hygiene practices, ensuring proper ventilation. Observe good personal hygiene, and do not eat, drink or smoke whilst handling.

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STORAGE CONDITIONS: Store in tightly closed original container, in a cool, dry & ventilated area away from heat sources & protected from light. Keep air contact to a minimum.

FIRE PROTECTION: Keep away from ignition sources & naked flames. Take precautions to avoid static discharges in working area.

9. EXPOSURE CONTROL/PERSONAL PROTECTION

Engineering Controls: Provide adequate ventilation. Safety showers and eye wash stations should be located near work areas.

Work Practice Controls:

Personal hygiene is an important work practice exposure control measure and the following general measures should be taken when working with or handling this material:

- (1) Do not store, use, and/or consume foods, beverages, tobacco products, or cosmetics in areas where this material is stored.
- (2) Wash hands and face carefully before eating, drinking, using tobacco, applying cosmetics, or using the toilet.
- (3) Wash exposed skin promptly to remove accidental splashes or contact with this material.

Respiratory: Avoid casual breathing of vapors

Ventilation: Local, mechanical exhaust in recommended

Eye: safety glasses or goggles

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Skin: Wear gloves and clothing to avoid skin contact

10. PHYSICAL AND CHEMICAL PROPERTIES

Physical State: Liquid, Oil

Color: Yellow to yellowish green liquid.

Odor: With the strong smell of eugenol.

Boiling: 251℃

Refractive Index: 1.527--1.535

Optical Rotation: -1° --+2°

Solubility in Water: Soluble in more than 70% ethanol.

Content: Eugenol $\geq$ 85.0

11. STABILITY AND REACTIVITY

Reactivity: It presents no significant reactivity hazards, by itself or in contact with water. Avoid contact with strong acids, alkali or oxidising agents.

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Decomposition: Liable to cause smoke & acrid fumes during combustion: carbon monoxide, carbon dioxide & other non-identified organic compounds may be formed.

12. TOXICOLOGICAL INFORMATION

CAUTION: Liquid may be irritating to eyes and skin. Not a sensitiser, not photo-toxic; however toxic to humans by oral route

13. ECOLOGICAL INFORMATION

Biodegradability: Data not available

Precautions: Prevent surface contamination of soil, ground & surface water.

14. DISPOSAL CONSIDERATIONS

Avoid disposing to drainage systems and into the environment. Seek expert advice.

15. TRANSPORT INFORMATION

Road (ADR/RID): N/A

AIR (IATA): N/A

SEA (IMDG): N/A

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Package: 180kg/Galvanized steel drum

16. ADDITIONAL INFORMATION

This information is provided for documentation purposes only.

The complete range of conditions or methods of use are beyond our control therefore we do not assume any responsibility and expressly disclaim any liability for any use of this product. Information contained herein is believed to be true and accurate. However , all statements or suggestions are made without warranty, expressed or implied, regarding accuracy of the information, the hazards connected with the use of the material or the results to be obtained from the use thereof.

This safety sheet cannot cover all possible situations which the user may experience during processing.

Each aspect of your operation should be examined to determine if, or where, additional precautions may be necessary. All health and safety information contained in this bulletin should be provided to your employees or customers.